AYANANTA CHOWDHURY

Designing documentation that AI systems can understand, reason over, and use

AI Documentation Architect | Knowledge Systems Architect

Location: India

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Professional Summary

AI Documentation Architect with **15+** years of experience in transforming traditional documentation into structured, machine-understandable knowledge, authoring technical documentation, and designing scalable knowledge systems for enterprise software. An industry thought leader in building semantic content maturity models, enabling the building of proactive, dynamic, and autonomous AI-native help systems (user copilots).

Work Experience

HCL Technologies | Senior Technical Manager – User Assistance (Mar 2023 – Apr 2025, Remote) Fulltime

Impact

- Built ontology-driven information architecture enabling future AI-assisted retrieval for a Saas ERP solution from SAP.
- Improved content reuse by 11% by implementing conkeyrefs & keyrefs in DITA-based structured authoring.

Key Contributions

- Audited content using TEXTML queries in IxiaSoft CCMS to identify duplicate content.
- Extracted domain entities using spaCy NLP pipelines.
- Normalized terminology using OpenRefine morphological variant detection.
- Built a glossary with canonical terms and recorded allowed and deprecated variants.
- Implemented Schematron validation rules to enforce terminology governance.
- Generated taxonomy facets using Carrot^2 clustering.
- Designed ontology models defining classes, attributes, and relationships.

Quincus | API Technical Writer (Oct 2022 – Nov 2022, Remote) Fulltime

- Explained API endpoints, resources, methods, authentication protocols, parameters, and headers (logistics domain).

SugarBox Networks | Technical Communications Manager (Aug 2020 – Sept 2022, Remote) Fulltime

Impact

- Designed an NLP-driven terminology extraction pipeline using spaCy to convert documentation into structured semantic entities supporting knowledge graph construction and AI retrieval workflows.

- Reduced CDN configuration time by 31.4%.

Key Contributions

- Built a controlled vocabulary and terminology governance framework.
- Designed a multi-faceted taxonomy and semantic metadata tagging strategy.
- Managed documentation backlog and content planning in JIRA.
- Measured documentation effectiveness using Google Analytics, Hotjar, and Algolia.

HCL Software | Technical Writing Senior Consultant (Nov 2018 – Mar 2020, Pune) Fulltime

Impact

- Streamlined the documentation workflow to align with the CI/CD release cycle for a data engineering platform.

Key Contributions

- Led and mentored the documentation team.
- Drafted Hotfix docs and release notes for a data-engineering solution and wrote a variety of UI texts (microcopy).
- Added parameter descriptions in the Open API Specification file and wrote the conceptual part of API documentation.

Sungard Discovery Services | Specialist, Technical Documentation (May 2017 - Jul 2017, Pune) Fulltime

- Produced video tutorials for the company's LMS platform, explaining the capabilities of a disaster recovery product.

Persistent Systems | Information Development Team Lead (Nov 2013 – May 2017, Pune) Fulltime

- Produced demo videos of medical device MVPs for potential investors (life sciences & healthcare domain).
- Coordinated with cross-functional teams of Merck & Co. for ITIL-based change management, using ServiceNow.
- Collaborated with the VMware on-site team to author the user and quick start guide for vRealize Network Insight.
- Ensured content followed WCAG 2.1 accessibility standards. Did usability testing to verify content accuracy.

Rolta India | Senior Technical Writer - Product Documentation (Aug 2012 – Jul 2013, Mumbai) Fulltime

- Authored content to explain new features developed for a GIS product using a single-sourcing tool, following Agile.

Polaris Software Lab | Technical Writer (Oct 2010 – Aug 2012, Mumbai) Fulltime

- Co-authored user and installation guides for a banking solution processing cross-border transactions via SWIFT messaging, complying with NACHA standards for ACH transfers, and ensuring accurate fund settlements.

Connectiva Systems | Technical Writer (Feb 2008 – Sept 2010, Kolkata) Fulltime

- Built storyboards for eLearning videos (telecom domain), edited corporate training PPTs, and updated the user guide.

Compare Infobase | Content Writer (Jan 2007– Dec 2007, Kolkata) Fulltime

- Wrote content on tourist attractions for travel websites following SEO guidelines.

Education

Postgraduate in Mass Communication and Journalism, Calcutta University | 2005–2006

Skills / Competencies

AI AND KNOWLEDGE SYSTEMS ARCHITECTURE

Design AI-ready documentation ecosystems that transform content into structured knowledge usable by LLMs.

- Extract named entities, create a glossary of canonical terms, and record their allowed and deprecated variants.
- Implement terminology governance using Schematron rules for consistent machine interpretation.
- Chunk content into atomic knowledge units, with each unit having a unique identifier.
- Design a multi-faced content taxonomy and design semantic models supporting AI copilots and RAG systems.
- Build formal semantic models using RDF, RDFS, OWL, and SKOS standards.
- Enable knowledge graph construction.
- Architect product documentation for hybrid retrieval architecture (vector search + knowledge graph reasoning).

CONTENT STRATEGY

Study and analyze the target audience to create user personas, journey maps, and workflow diagrams. Define KPIs for successful user assistance, measure help success, create a master documentation list with timeliness, and mentor teams.

CONTENT DEVELOPMENT

Create multimodal help content enabling users to set up, operate, and maintain enterprise applications through video tutorials, guides, and visual learning assets such as diagrams, infographics, and workflows. Author release notes, hotfix doc, FAQs & troubleshooting, and training materials. Produce SEO-optimized (E-E-A-T) marketing content and refine SOPs, runbooks, process documentation, and functional & technical specifications in alignment with MSTP guidelines. Embed assistance in the form of microcopy (UX writing) and walkthrough videos (digital guided tours).

API DOCUMENTATION

Deliver exceptional developer experiences through well-structured API documentation with interactive examples.

LANGUAGE

JSON, XML, YAML, Markdown, HTML (working knowledge), Python (only familiarity)

Tools

Information Architecture: IxiaSoft, SpaCy, Open Refine, Carrot^2, Protégé | **Docs-as-Code**: Git, Markdown | **Audio-Visuals & Diagramming**: Camtasia, Visio, Napkin, HeyGen, and Lucidchart | **Analytics**: Google Analytics, Hotjar, and Algolia | **Authoring**: DITA XML Editors (Oxygen, Arbortext), RoboHelp / MadCap Flare, Confluence, Docs, MS Office | **DAP**: WalkMe | Project Management/Ticketing ITSM: JIRA, ServiceNow | **API Documentation**: Swagger, Postman | **Vibe Coding**: Cursor AI

Certifications

DITA XML Authoring | UX Writing | API Documentation | Camtasia | Git